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System and method for assessing device usage

Abstract

A method for monitoring device usage includes: receiving user device information; based on the user interaction data, determining a state associated with the user device; and based on the state of the user device, determining a set of tap parameters. Additionally or alternatively, the method can include any or all of: determining a risk score based on at least on the set of tap parameters; determining an output based on the risk score; and/or any other suitable processes performed in any suitable order.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 16/700,991, filed 2 Dec. 2019, which is incorporated in its entirety by this reference.

TECHNICAL FIELD

(1) This invention relates generally to the vehicular activity monitoring field, and more specifically to a new and useful system and method for monitoring device usage by a user in the vehicle activity monitoring field.

BACKGROUND

(2) A user's driving behavior, such as the riskiness of their driving behavior, is dependent upon many factors. A large factor in recent years has been device usage while driving, such as texting, calling, engaging an application, and various other behaviors. While some conventional systems and methods can approximate that a mobile device is in a vehicle, there are various different ways in which the user can interact with the device, each of which can be associated with vastly different levels of risk.

(3) Thus, there is a need in the vehicle monitoring field to create a new and useful system and method for assessing and properly classifying mobile device usage associated with a driver of a vehicle.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a schematic representation of a method **100** for assessing device usage.
- (2) FIG. 2 is a schematic representation of a system **200** for assessing device usage.
- (3) FIGS. 3A-3C depict a variation detecting user interaction with a secondary client application.
- (4) FIGS. 4A-4C depict a variation of different types of device usage.
- (5) FIG. 5 depicts a variation of a method **100** for assessing device usage.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(6) The following description of the preferred embodiments of the invention is not intended to limit the invention to these preferred embodiments, but rather to enable any person skilled in the art to make and use this invention.

1. Overview

(7) As shown in FIG. 1, a method **100** for monitoring device usage includes: receiving user device information **S110**; based on the user interaction data, determining a state associated with the user device **S120**; and based on the state of the user device, determining a set of tap parameters and/or a device classification **S130**. Additionally or alternatively, the method **100** can include any or all of: determining a risk score **S140**; determining an output based on the risk score **S150**; and/or any other suitable processes performed in any suitable order.

(8) As shown in FIG. 2, a system **200** for monitoring device usage includes: a client application executing on a user device. Additionally or alternatively, the system **200** can include and/or be configured to interface with any or all of: a set of sensors, a user device, a processing system, a computing system (e.g., remote computing system), a vehicle, and/or any other suitable component(s).

2. Benefits

(9) Variations of the system and method can confer various benefits over existing systems and methods.

(10) First, in some variations, the system and/or method confers the benefit of distinguishing between various types of device usage by a driver, each of which can be associated with different levels of risk. In specific examples, the method confers the benefit of enabling a determination to be made of whether the user is engaging with a mobile device while the mobile device is being held by a user or while the mobile device is in a mount/holder (e.g., mounted to a dashboard of the vehicle). This information can be used by an insurance company to appropriately assess the riskiness of a driver. In other specific examples, additional or alternative to these, the system and/or method can be used to provide feedback (e.g., notification of time spent in a state of distracted driving, risk score, etc.) and/or another output (e.g., suspension of device usage while driving) to the driver and/or another suitable user.

(11) Second, in some variations, the system and/or method confers the benefit of minimizing a number of false positives associated with the driver engaging in device usage behavior. In specific examples, for instance, the system and/or method, in assessing device usage, takes into account a central processing unit (CPU) spiking for reasons (e.g., downloading data, transitioning between songs in an audio application, etc.) unrelated to device usage by the user.

(12) Third, in some variations, the system and/or method confers the benefit of assessing device usage associated with a mobile device based only on data collected at the mobile device. In specific examples, for instance, any or all of motion data, processing data, and proximity/contact data can be used to determine device usage.

(13) Fourth, in some variations (e.g., as shown in FIGS. 3A-3C), the system and/or method confers the benefit of assessing a user's interaction at a secondary client application with a primary client application, wherein information is not directly received at the primary client application from the

secondary client application. In specific examples, for instance, a telematics client application can indirectly assess usage of a secondary client application through the detection and processing of any or all of: motion information of the user device (e.g., from an accelerometer, from a gyroscope, etc.), orientation information of the user device (e.g., from a gyroscope, from an orientation sensor, etc.), proximity information (e.g., from a proximity sensor of the user device), processing activity of the user device (e.g., CPU activity, network transfer information, available free memory, etc.), and/or any other suitable sensor information.

(14) Fifth, in some variations, the technology confers the benefit of enabling the calculation of a set of one or more insurance costs for the user based on a calculated risk score for the driver, which can be used, for instance, to guide, recruit, and/or offer insurance policies to the user according to these insurance costs. Additionally, the technology can further confer the benefit of facilitating communications and/or interfaces and/or information exchanges between multiple entities involved in the implementation of presenting insurance costs and/or offers to drivers commensurate with their driving risk scores. In a set of examples, for instance, the technology confers the benefit(s) of: decreasing the cost(s) and/or burden for insurance providers associated with recruiting new and/or good drivers; providing drivers with the best insurance rates based on their actual driving behaviors (e.g., rather than purely demographic information); providing a valuable benefit to 3rd party application entities (e.g., separate from the insurance entity) who provide their users with the opportunity to increase their insurance savings (e.g., by facilitating the matching of the driver with a lower-cost insurance entity); and/or any other benefits.

(15) Additionally or alternatively, the system and/or method can confer any other suitable benefit(s).

3. System

(16) The system **200** functions to provide information with which to assess a device usage by a user. Additionally or alternatively, the system **200** can function to perform at least a portion of the processing required to determine a device usage, transfer information to another entity (e.g., remote computing system) at which the processing is performed, produce an output, and/or perform any other suitable function.

(17) The system **200** can include and/or be configured to interface with a user device, wherein the device usage is associated with the user device. Examples of the user device include a tablet, smartphone, mobile phone, laptop, watch, wearable device (e.g., glasses), or any other suitable user device. The user device can include power storage (e.g., a battery), processing systems (e.g., CPU, GPU, memory, etc.), user outputs (e.g., display, speaker, vibration mechanism, etc.), user inputs (e.g., a keyboard, touchscreen, microphone, etc.), a location system (e.g., a GPS system), sensors (e.g., optical sensors, such as light sensors and cameras, orientation sensors, such as accelerometers, gyroscopes, and altimeters, audio sensors, such as microphones, magnetometers, etc.), data communication system (e.g., a WiFi transceiver(s), Bluetooth transceiver(s), cellular transceiver(s), etc.), or any other suitable component.

(18) The system **200** can additionally include and/or be configured to interface with a set of sensors (e.g., as part of user device, separate from user device, etc.), which can include any or all of: motion sensors (e.g., accelerometer, gyroscope, magnetometer, orientation sensor, etc.), which can function to detect any or all of: user device movement, user device orientation, vehicle movement, vehicle orientation, position of user device within the vehicle, and/or any other suitable information; proximity sensors (e.g., optical sensors, capacitive sensors, etc.), which can function to detect and/or classify a user's handling of a user device; location sensors (e.g., GPS); any or all of the sensors described above; any or all of the sensors described below; and/or any other suitable sensors.

(19) In addition to or alternative to the information described below in the method **100**, the sensor system can collect any or all of the following user activity data and vehicle data. User activity data (e.g., user activity datasets) can include any one or more of: device datasets (e.g., describing

devices associated with the user; historical pairings and/or communications with other devices, such as vehicles and/or other user devices via Bluetooth or other communication protocols; etc.); behavioral datasets (e.g., behavior data upon which user activity models can be based or generated for users and/or user accounts; social media datasets derived from social media actions taken by the user and/or related to the user, which can indicate user activity states corresponding to social media actions; physical activity datasets associated with movement patterns of the user; etc.); demographic datasets (e.g., nationality, ethnicity, age, gender, etc.); device event datasets (e.g., associated with text messaging; phone calling; device idling; device charging; application usage; sensor usage; intelligent personal assistants; outputs such as media casting; inputs; etc.), and/or any other suitable datasets associated with a user. Vehicle data can include any one or more of: proximity sensor data (e.g., radar, electromagnetic sensor data, ultrasonic sensor data, light detection and ranging, light amplification for detection and ranging, line laser scanner, laser detection and ranging, airborne laser swath mapping, laser altimetry, sonar, etc.), vehicle camera data (e.g., in-car cameras, exterior cameras, back-up cameras, dashboard cameras, front-view cameras, side-view cameras, image recognition data, infrared camera, 3D stereo camera, monocular camera, etc.), car speed, RPM data, odometer, altimeter, location sensor data (e.g., GPS data, compass data, etc.), motion sensor data (e.g., from an accelerometer, gyroscope, etc.), environmental data (e.g., pressure, temperature, etc.), light sensor data (e.g., from infrared sensors, ambient light sensors, etc.), fuel level (e.g., percentile-based, distance-based, low warning, etc.), fuel system status, oxygen sensor data, throttle position, gear data (e.g., drive, neutral, park, reverse, gear number, etc.), HVAC data (e.g., current temperature, target temperature, etc.), driving status (e.g., restricted features such as user inputs into control panel, unrestricted features, etc.), connectedness status (e.g., near-field communication pairing between the vehicle and a user's mobile device), and/or any other suitable vehicle data. For example, vehicle data can be used in comparisons with mobile device-derived movement data in training, updating, and/or otherwise processing computational models described herein and/or other suitable models. However, supplementary data types can be configured in any suitable manner.

(20) The system **200** can include one or more client applications executing on the user device, which can function to collect information with which to determine a device usage and/or perform any or all of the processing required to determine the device usage.

(21) The client applications can include a primary client application, which is preferably a telematics application, which functions to receive information from one or more sensors (e.g., as described in **S110**). Additionally or alternatively, the primary client application can perform any or all of the process required to determine a device usage.

(22) The client applications can optionally include one or more secondary client applications (e.g., messaging applications, native user device applications, social media applications, viewer applications, Facebook, YouTube, etc.), wherein the primary client application indirectly detects usage of (e.g., without directly receiving data from) the secondary client applications (e.g., as shown in **FIGS. 3A-3C**).

(23) Additionally or alternatively, the client applications can be otherwise arranged or described.

(24) The system **200** preferably interfaces with a vehicle, wherein the device usage is determined for a user driving the vehicle and to which the user device (equivalently referred to herein as a mobile device) is optionally removably coupled (e.g., at a mount).

(25) The vehicle to which the mobile electronic device is removably coupled (e.g., present within, mounted to an interior of, etc.), and in the context of which variations of the method **100** can be implemented, can be an automotive vehicle (e.g., a car, truck, SUV, etc.), a light electric vehicle (e.g., an electric bike, an electric scooter, an electric skateboard, any other suitable electric vehicle that is configured to carry at least one human, etc.), an aerial vehicle (e.g., a fixed wing aircraft, a rotary wing aircraft, a drone, a hovercraft, etc.), a mass transit land vehicle (e.g., a bus, a locomotive train, a light-rail train, etc.), an aquatic vehicle (e.g., a pleasure craft, a ferry, a

hydrofoil craft, etc.), and/or any other suitable vehicle.

(26) Additionally or alternatively, the system **200** can include any other suitable components and/or be otherwise configured.

4. Method

(27) The method **100** functions to assess the usage of a mobile device by a user driving a vehicle. Additionally or alternatively, the method **100** can function to distinguish between multiple types of device usage (e.g., phone call vs. texting vs. interacting with a phone in a mount); quantify a device usage (e.g., duration, severity, etc.); determine a risk score for a user associated with the riskiness of their driving; produce an output based on the device usage and/or risk score (e.g., report to an insurance company, limit device usage, provide awareness to the user, etc.), and/or perform any other suitable function.

(28) The method **100** is preferably performed with a system as described above, but can additionally or alternatively be performed with any other suitable system.

(29) 4.1 Method: Receiving User Device Information **S110**

(30) The method **100** includes receiving user device information **S110**, which functions to receive information with which to determine a device usage associated with a mobile device by the user. Additionally or alternatively, **S110** can function to eliminate data to be used when determining a device usage (e.g., to reduce false positives, to remove noise, etc.) and/or perform any other suitable function(s).

(31) The user device information, equivalently referred to herein as user device data, is preferably generated and/or sampled at the user device, such as at any or all of: one or more sensors of the sensor system, at a processing system (e.g., CPU) of the user device, at a computing system of the user device, at a client application executing on the user device, at an SDK associated with a client application, and/or at any other suitable component(s) of the user device. Information received from a user device is preferably received from a single user device, but can additionally or alternatively be received from multiple user devices.

(32) Additionally or alternatively, information can be received from any or all of: an external sensor system; a vehicle (e.g., speedometer); a data storage source; a remote computing system; and/or any other suitable data source.

(33) The information is preferably received at a user device (e.g., the same user device as where it's generated, a separate user device, etc.), further preferably at a client application (e.g., primary client application, secondary client application, etc.) executing on the user device, but can additionally or alternatively be received at any or all of: a processing system of the user device, a computing system of the user device, an SDK associated with the client application, and/or at any other suitable component or module of the user device. Further additionally or alternatively, information can be received at a remote computing system and/or any other suitable component(s).

(34) The information collected in **S110** (and/or supplementary datasets) is preferably sampled by components arranged at a mobile device, but can additionally or alternatively be sampled by components associated with (e.g., arranged at, positioned within, mounted to, physically connected to, etc.) any suitable device and/or vehicle. The user device information (and/or supplementary datasets) is preferably associated with one or more users (e.g., collected at a mobile device of the user; collected at a mobile device physically proximal a user; stored in association with user accounts for one or more users; movement datasets describing user movement, such as pre-, during, and/or post-driving session; etc.). Further, the user device information is preferably associated with one or more driving sessions, which can be characterized by characteristics and/or patterns in movement data (e.g., movement data indicating movement that is characteristic of a vehicle) and/or supplementary data (e.g., engine sounds as sensed by a smartphone microphone), and where driving sessions can include movement of, stoppage of, operation control related to, and/or other suitable aspects associated with vehicles. Additionally or alternatively, the user device information can be associated or independent of any suitable entities.

(35) Any or all of the information in **S110** can be collected continuously from and/or at mobile devices in a vehicle (e.g., during vehicle operation), such as in real-time, but can additionally or alternatively include collecting information periodically (e.g., at vehicle stoppage events, at predetermined time intervals such as every minute, etc.), and/or in temporal relation (e.g., in response to, etc.) to trigger conditions (e.g., movement-related thresholds; satisfaction or failure of device association conditions; detection of user's interaction with the user device; conditions associated with supplementary data; etc.), but collecting information can be performed at any suitable time and/or frequency. Any or all of the information can be associated with one or more temporal indicators (e.g., a time period, a time point, etc.) describing when the information was collected (e.g., sampled at the corresponding sensor, etc.), determined, and/or otherwise processed. Additionally or alternatively, associated temporal indicators can describe temporal context surrounding the information but can further additionally or alternatively correspond to any suitable temporal indicator or be time-independent. However, collecting information can be performed in any suitable manner.

(36) **S110** can be performed by any or all of: an application (e.g., a native application executing in the background, a client, etc.); the operating system of the mobile device (e.g., iOS™, Android™); a remote computing system associated with the operating system; or by any suitable system. In variants where **S110** is performed by the operating system or remote system associated with the operating system, the method can reduce additional resource consumption (e.g., battery consumption) because the underlying data is being collected and used for other processes; the application itself is minimizing or not consuming any additional resources. In one variation, **S110** includes setting a wake condition with the mobile device operating system, wherein the mobile device operating system delivers an event (e.g., notification) to the application, launches the application, progresses to **S120**, or otherwise triggers further application operation in response to the condition being met.

(37) The user device information preferably includes movement information (equivalently referred to herein as motion information), which is preferably associated with motion of the user device (e.g., with respect to the vehicle), but can additionally or alternatively be associated with the vehicle (e.g., speed of vehicle), the user, and/or any other suitable component(s).

(38) The movement information can include any or all of: movement data (e.g., position data; velocity data; acceleration data; a position, velocity, acceleration (PVA) dataset; a motion dataset, motion data, movement data, etc.); orientation data (e.g., orientation of the user device, change in orientation direction at which a screen of the user device is facing, gravity deviation, rate of gravity deviation, rate of gravity deviation around a projected centroid, etc.); and/or any other suitable information.

(39) The movement information is preferably sampled at one or more movement sensors indicative of motion and/or orientation, such as any or all of: multi-axis accelerometers, single-axis accelerometers, gyroscopes, and/or any other motion or orientation sensors (e.g., virtual sensor utilizing accelerometer and gyroscope data; virtual sensor utilizing accelerometer, gyroscope, and magnetometer data; etc.). Additionally or alternatively, motion information and/or any other information collected in **S110** can be collected from and/or derived from any or all of: location sensors (e.g., GPS data collection components, magnetometer, compass, altimeter, etc.), optical sensors, audio sensors, electromagnetic (EM)-related sensors (e.g., radar, lidar, sonar, ultrasound, infrared radiation, magnetic positioning, etc.), environmental sensors (e.g., temperature sensors, altitude sensors, pressure sensors, etc.), biometric sensors, and/or any other suitable sensors.

(40) In some variations, the motion information includes acceleration data collected from an accelerometer of the user device and orientation data (e.g., rate of gravity deviation) collected from a gyroscope and/or an orientation sensor (e.g., virtual sensor) of the user device (e.g., which can be aggregated through sensor fusion into a gravity deviation parameter). The motion information is preferably collected continuously while the user is driving (e.g., as detected by the accelerometer,

as detected by a GPS application, etc.), but can additionally or alternatively be collected in response to another trigger, collected intermittently, and/or otherwise collected.

(41) In a set of examples, for instance, a gravity sensor in the form of a software sensor is utilized which aggregates (e.g., fuses) data from multiple motion and/or orientation sensors (e.g., accelerometers and gyroscope) to produce a gravity deviation parameter which represents the extent to which the user device deviates in orientation from the direction of gravity. The gravity deviation parameter preferably includes a set of one or more angle values (e.g., 3 angle values to represent a 3-dimensional deviation from the direction (e.g., axis) of gravity, but can additionally or alternatively include a single angle value, an aggregated angle value, values other than angle values, and/or any other parameters.

(42) Further additionally or alternatively, **S110** can be performed in absence of collecting motion information.

(43) The user device information further preferably includes system performance information, which functions to determine whether or not the user is engaging with the user device. The system performance information can additionally or alternatively function to determine if one or more secondary client applications are being used, determine which client applications are being used, quantify (e.g., in duration of time, number of interactions, etc.) the user's engagement with the user device, and/or perform any other suitable function.

(44) The system performance information preferably includes processing information associated with the user device, such as activity information collected at the CPU of the user device, which can be associated with any or all of: network transfer activities, downloading activities, and/or any other suitable activities associated with one or more processing systems of the user device.

(45) The system performance information can additionally or alternatively include any other suitable information associated with the user device, such as an amount of free memory.

(46) In some variations, the system performance information enables a primary client application to detect that a user is interacting with a secondary client application, despite not having direct access to the information being collected at the secondary client application. In specific examples, this is enabled by collecting CPU activity information, network transfer information, and an amount of free memory (e.g., which can be compared with a previous value to determine and/or predict a user's activity of the device) associated with the user device.

(47) Additionally or alternatively, any other suitable information can be collected and/or **S110** can be performed in absence of collecting system performance information.

(48) The device information further preferably includes device handling information, which functions to determine if and how the user is physically interacting with (e.g., holding in hand for texting, holding close to ear for calling, etc.) the user device. Collecting device handling information can additionally or alternatively function to determine if the phone is not being held by the user (e.g., is held in a mount, is otherwise coupled to an interior of the vehicle, is resting on a seat of the vehicle, etc.), determine a degree to which the user is interacting with the user device, determine and/or predict an activity (e.g., texting, calling, viewing a map and minimally interacting, etc.) associated with the interaction, and/or perform any other suitable function(s).

(49) The handling information preferably includes proximity and/or contact information, which can be collected at any or all of: proximity sensors (e.g., infrared sensors), optical sensors (e.g., light sensor, ambient light sensors, etc.), contact sensors (e.g., electrical conductance sensor), heat sensors, and/or any other suitable sensors configured to determine if the user device is proximal to the user (e.g., touching, proximal, proximal to the user's face, etc.). The handling information is preferably collected by one or more sensors onboard the user device (e.g., a proximity sensor proximal to a speaker of the user device, an ambient light sensor proximal to a speaker of the user device, etc.), but can additionally or alternatively be collected at any other suitable sensors onboard or external to the user device.

(50) Data from the proximity sensor preferably includes return signals and/or a return field

resulting from the emission of an electromagnetic field and/or a beam of electromagnetic radiation (e.g., infrared radiation). This can be used for instance, to determine if the user device is being held to the user's ear (e.g., non-stationary and proximity sensor indicates user is close, during a phone call, etc.) versus being held in a user's hand (e.g., non-stationary and proximity sensor does not indicate that a user is close, while texting, etc.). Data from another sensor, such as an optical sensor (e.g., ambient light sensor), can additionally or alternatively be used (e.g., to provide a higher confidence determination of how the user is interacting with the phone, to detect a user speaking on speaker phone and holding the phone backward, to distinguish between a user holding the phone and the phone resting face down on a seat, etc.). The second sensor can function, for instance, to distinguish between different types of handling of the user device by the user. Additionally or alternatively, any other suitable sensors can be used.

(51) The sensors can be arranged proximal to each other (e.g., on the same surface of the user device), on different surfaces of the user device (e.g., to accurately characterize a user's handling of the phone, etc.), all onboard the user device, all offboard the user device, and/or any combination of these.

(52) In some variations, the handling information is collected at a proximity sensor arranged onboard the user device (e.g., proximal to a speaker, proximal to a screen, etc.) and at an ambient light sensor arranged onboard the user device (e.g., proximal to a speaker, proximal to a screen, etc.), which are collectively used to determine and/or predict how the user is handling a non-stationary phone (e.g., holding up to ear, holding in hand, etc.).

(53) Additionally or alternatively, any other suitable information can be collected and/or **S110** can be performed in absence of collecting handling information.

(54) The method **100** can additionally include collecting location information **S112**, which functions to determine one or more characteristics of the vehicle. The location information can additionally or alternatively function to characterize and/or eliminate user device information based on one or more parameters (e.g., speed) associated with the vehicle, and/or perform any other suitable function.

(55) **S112** can include collecting one or more location datasets (e.g., collecting location data), which can include any one or more of: GPS data (e.g., position coordinates, associated time stamps, etc.), indoor positioning system data, local positioning system data, multilateration data, GSM localization data, self-reported positioning, control plane locating data, compass data, magnetometer data, and/or any other suitable location-related data. In an example, GPS data can be leveraged for complete PVA solutions, but can additionally or alternatively include any movement data, such as retrieved using GNSS data (e.g., via GLONASS, Galileo, BeiDou). In another example, proximity sensors associated with mobile phones (e.g., for capacitive proximity sensing; IR-based proximity sensing; etc.) can be used to in determining location of objects (e.g., users within a vehicle). In another example, **S112** can include collecting a micro-location dataset (e.g., sampled using beacons within and/or mounted to a vehicle). In another example, **S112** can include collecting a location dataset sampled by a mobile device and indicating the location of the mobile device within the vehicle (e.g., proximal front of vehicle; back of vehicle; etc.). However, collecting location datasets can be performed in any suitable manner.

(56) In some variations, **S112** includes continuously collecting location data of the vehicle, which can be used to determine any or all of: a speed of the vehicle, a road being traveled by the vehicle, one or more turns of the vehicle, and/or any other suitable information.

(57) In one variation, **S110** includes continuously collecting (e.g., while the user device is in a vehicle, while the user device is in a vehicle as determined by location information, etc.) motion information (e.g., from an accelerometer, from a gyroscope, from an orientation sensor, etc.) associated with the user device, performance information (e.g., processing activity, CPU activity, network transfer information, amount of free memory, etc.) associated with the user device, and optionally handling information (e.g., from a proximity sensor, from an optical sensor, etc.)

associated with the user device, which is used to assess a user's interaction with the user device in subsequent processes of the method. Additionally, location information can be collected and used to determine one or more parameters of the vehicle (e.g., speed, location in space, etc.), which can also be used in subsequent processes of the method (e.g., to eliminate data, to appropriately determine a risk score, etc.).

(58) Additionally or alternatively, any other suitable information can be collected in **S110**.

(59) 4.2 Method: Based on the User Device Information, Determining a State of the User Device **S120**

(60) The method **100** includes determining a state of the user device **S120**, which functions to classify whether or not the user device is stationary with respect to the vehicle. Additionally or alternatively, **S210** can function to determine any other characterization of the user device, such as any or all of: the user device is being handled (e.g., held, moved around, etc.) by the user, the user device is coupled to the vehicle such as in a mount, and/or any other suitable characterization.

(61) The state preferably indicates whether or not the user device is stationary (e.g., with respect to the vehicle) based on a stationary metric, which is preferably determined, at least in part, based on any or all of the motion information (e.g., acceleration information, gyroscope information, orientation information, etc.) described above. Additionally, the state can be determined based on any or all of the other information described above (e.g., performance information, handling information, location information, etc.), as well as any other supplementary data (e.g., environmental data). The stationary metric can include a binary indication of whether or not the device is stationary; a classification of the state as any or all of: stationary, non-stationary, in transition between stationary and non-stationary; a parameter along a spectrum of how stationary the device is; and/or any other suitable metric(s). The stationary metric can be determined based on any or all of: one or more algorithms, a machine learning model (e.g., a deep learning model), comparison with a threshold, and/or determined in any other suitable way.

(62) The state of the user device preferably serves as a trigger for one or more sub-processes of **S130** (e.g., determining whether phone is handled by a user or in a mount, etc.), but can additionally or alternatively be used in any suitable way(s) throughout the method **100**.

(63) The state of the device can be determined for each of a set of sections of data, equivalently referred to herein as segments of data, wherein the sections of data are separated by a transition (e.g., a motion threshold, a performance threshold, a device handling threshold, etc.), and/or any other suitable information.

(64) In one variation, **S120** includes classifying, based on any or all of the information collected in **S110** and an algorithm (e.g., of a machine learning model), a classification of at least whether the device is in a stationary state or a non-stationary state, which functions to trigger one or more actions in **S130**.

(65) 4.3 Method: Based on the State of the User Device, Determining a Set of Tap Parameters and/or a Device Classification **S130**

(66) The method **100** includes based on the state of the user device, determining a set of tap parameters and/or a device classification **S130**, which functions to determine (e.g., quantify, assess, etc.) a user's (e.g., driver's) interaction with the user device.

(67) Taps herein refer to any actions or gestures associated with a user's interaction with a user device, further preferably with a touch-sensitive surface of the user device, such as, but not limited to: a tap or touch, a swiping motion, a pinching motion, a zooming motion (e.g., zooming in, zooming out, etc.), a typing action, and/or any other suitable interaction. Additionally or alternatively, taps can refer to interactions with other components of the user device, such as a pressing of a button (e.g., power, volume, etc.), a user of one or more sensors (e.g., an audio sensor to provide a command, etc.), and/or any other suitable interactions.

(68) The tap parameters can include and/or be determined based on any or all of: a number of taps, a frequency of taps, a duration of taps, a time in between taps, a set of taps (e.g., closely-spaced

taps), a tap region (e.g., an aggregated set of taps), a tap event (e.g., which remains after comparisons with a predetermined set of thresholds), and/or any other suitable parameters.

(69) In an event that the device is in a stationary state (e.g., during a section of data collected in **S110**), **S120** preferably includes determining a first set of tap parameters **S132**.

(70) Determining that the user device is in a stationary state preferably triggers a second state to be determined, which determines whether or not the user device is in a mount. The second state can be determined based on any or all of: an algorithm, a deep learning model, comparison with a threshold, sensor data (e.g., visual data from a camera, etc.), any of the information collected in **S110**, knowing the location of the user device within the vehicle (e.g., consistently at the front of the vehicle, on a dashboard, etc.) and/or any other suitable information.

(71) In an event that it is determined that the user device is in a mount, **S132** preferably includes determining (e.g., computing) a set of tap parameters, wherein the set of tap parameters are determined, at least in part based on motion information collected in **S110**, such as data collected from any or all of: an accelerometer, a gyroscope, and an orientation sensor. Additionally or alternatively, the set of tap parameters can be based on any other suitable sensor information and/or any other information.

(72) The set of tap parameters preferably includes a set of tap regions, which can be determined based on any or all of the following processes. Additionally or alternatively, the set of tap parameters can include any or all of the intermediate parameters described below, any or all of the parameters described in **S134**, and/or any other suitable parameters.

(73) In preferred variation, the set of tap parameters is determined based on a gravity deviation parameter, which can reflect a user's interaction with (e.g., perturbation of) the user device (e.g., tapping the screen, asymmetrically tapping the screen, causing perturbations to the phone while it is mounted, etc.) while the device is mounted. The gravity deviation parameter is preferably calculated about a projected centroid (e.g., medoid) of the user device, but can additionally or alternatively be calculated about any suitable region, point, axis, plane, or other feature (physical or virtual) of the user device. In specific examples, the gravity deviation parameter includes a rate of gravity deviation. Additionally or alternatively, any other gravity deviation parameter can be calculated and/or any other suitable parameter can be calculated.

(74) **S132** can optionally include classifying at least a subset of the gravity parameters as potential tap events, which functions to eliminate tap events which are likely not associated with any or all of: the user interacting with the user device (e.g., and instead noise), the user being distracted by his interaction with the user device (e.g., for brief or rare taps), and/or can perform any other suitable function. In some variations, a predetermined percentage (e.g., 10%, less than 10%, greater than 10%, between 2% and 25%, etc.) of highest-value (e.g., greatest rate of gravity deviation) tap parameters are classified as potential tap events. The predetermined percentage can be determined. Additionally or alternatively, tap parameters which exceed a predetermined threshold (e.g., minimum rate of gravity deviation threshold) can be classified as potential tap events, a set of random tap parameters can be classified as potential tap events, all random tap parameters can be classified as potential tap events, and/or potential tap events can be determined in any other suitable way(s).

(75) **S132** can optionally include aggregating the set of potential tap events into tap regions, which functions to take into account one or more temporal parameters (e.g., duration of tap event, duration of a set of tap events, time between subsequent tap events, etc.) associated with the user's interaction with the user device. Additionally, this can function to eliminate events not associated with a user interaction (e.g., a false positive), not significant in causing user distraction, and/or any other suitable events. The temporal parameter preferably includes a time between adjacent tap events, which is compared with a maximum time threshold (e.g., 10 seconds, less than 10 seconds, greater than 10 seconds, etc.), wherein potential tap events which have a time spacing below the time threshold are aggregated into a tap region. Additionally or alternatively, a deep learning model

and/or algorithm can be used to determine a tap region, a minimum threshold can be used to determine a tap region, a set of multiple thresholds can be used, and/or any other suitable parameters can be used to determine a tap region. Further additionally or alternatively, **S132** can be performed in absence of aggregating potential tap events into a tap region. In some variations, a set of potential tap events which have a time between events within 10 seconds form a tap region.

(76) **S132** can optionally include discarding one or more tap regions based on a predetermined threshold, such as a temporal threshold (e.g., duration of tap region, number of potential tap events in the tap region, etc.). In some variations, **S132** includes discarding tap regions (e.g., removing from further consideration) having a duration (e.g., from the beginning of the first potential tap event to the end of the last potential tap event) of less than a minimum threshold (e.g., 3 seconds, less than 3 seconds, greater than 3 seconds, etc.).

(77) **S132** can optionally include analyzing the tap regions, which functions to determine one or more features associated with the tap region. These features can be used to eliminate tap regions from further consideration, rank a riskiness of tap regions, and/or perform any other suitable function. The tap regions are preferably analyzed with respect to system performance information, further preferably with respect to processing activity (e.g., CPU usage) of the user device associated with the tap region. This can function, for instance, to further narrow in the tap regions which are most likely associated with user interaction with the user device. In preferred variations, for instance, **S132** includes analyzing each of the set of tap regions for an associated CPU usage occurring contemporaneously (e.g., at the same time, substantially at the same time, etc.) with the tap region, wherein tap regions associated with a CPU usage below one or more predetermined thresholds are discarded (e.g., removed, eliminated from further consideration, etc.). In specific examples, a CPU curve (e.g., scaled, normalized, smoothed, scaled and smoothed, etc.) associated with each tap region is analyzed, and if the CPU curve has a predetermined number of points (e.g., 2, 1, greater than 2, etc.) above a predetermined threshold value (e.g., 0.75, 0.5, less than 0.75, greater than 0.75, etc.), it moves forward to further processing; otherwise it is discarded. Additionally or alternatively, one or more thresholds can be dynamically determined (e.g., based on the collective set of CPU curves associated with the tap region), a different number of thresholds can be implemented, an algorithm and/or a deep learning model can be implemented, and/or the tap regions can be otherwise analyzed and/or processed.

(78) **S132** can optionally include discarding and/or chopping one or more tap regions based on a vehicle parameter (e.g., as determined based on information collected in **S110**, as determined based on location information, etc.), such as a vehicle speed, which can function to discard tap regions associated with low vehicle speeds (e.g., less risky behavior) and/or no speed. In some variations, for instance, **S132** includes discarding and/or chopping (e.g., reducing a duration of) tap regions associated with a vehicle speed below a predetermined threshold (e.g., 5 meters per second, less than 5 meters per second, greater than 5 meters per second, 11 miles per hour, less than 11 miles per hour, greater than 11 miles per hour, etc.). Additionally or alternatively, the tap regions can be otherwise analyzed.

(79) In an event that the device is not in a stationary state and is rather in a mobile state, **S120** preferably includes determining a classification of the user device **S134**, wherein the classification is determined, at least in part, based on movement information (e.g., orientation information, motion information, etc.) collected in **S110**, such as data collected from any or all of: an accelerometer, a gyroscope, and an orientation sensor. Additionally or alternatively, the classification can be based on any other suitable sensor information and/or any other information. Further additionally or alternatively, **S134** can include determining a second set of tap parameters (e.g., similar to those determined in **S132**, identical to those determined in **S132**, differently from those determined in **S132**, based on movement data, based on processing activity data, etc.), and/or any other suitable parameters. The second set of tap parameters can include a set of tap sections, which can be determined based on any or all of the following processes. Additionally or

alternatively, the second set of tap parameters can include any or all of the intermediate parameters described below, any or all of the parameters described in **S132**, and/or any other suitable parameters.

(80) **S134** preferably includes splitting any or all of the user interaction data (e.g., any or all of the data collected in Silo) into sections based on a set of transitions, wherein the set of transitions preferably separate stationary events from mobile events (e.g., based on motion information), but can additionally or alternatively separate different types or severities (e.g., in terms of risk) of mobile events, and/or otherwise determine subsections of data. Further additionally or alternatively, **S134** can be performed based on a collective set of sections.

(81) **S134** further preferably includes classifying each section of user interaction data, which can function to determine and/or approximate how the user is interacting with the user device (e.g., as shown in FIGS. 4A-4C). The sections are preferably determined, at least in part, based on orientation information (e.g., from a gyroscope of the user device, from an orientation sensor of the user device, from an accelerometer of the user device) associated with the user device, wherein the orientation information can be used to determine and/or approximate if the user is engaging in a phone call; if the user is interacting with a screen of the user device (e.g., reading, texting, interacting with a client application, providing inputs to a client application, receiving outputs from a client application, etc.); and/or any other suitable classification (e.g., unknown interaction, no interaction, etc.). In some variations, for instance, an orientation corresponding to the screen facing upward or nearly upward corresponds to a user interacting with the screen of the user device. An orientation corresponding to a screen facing to the side or substantially to the side (e.g., against a user's ear) can correspond to a user making a phone call. Additionally or alternatively, the orientation information can be otherwise used in producing any or all of the classifications.

(82) Additionally or alternatively, any or all of the proximity information can be used in classifying each section of user interaction data. In some variations, for instance, the proximity information can indicate whether a user device is being held against the user (e.g., against the user's ear), or if the user device is not arranged proximal to the user (e.g., being held in a user's hand with the screen facing up, not being held by the user, etc.).

(83) Further additionally or alternatively, any other information (e.g., collected in **S110**, outside of **S110**, etc.) can be used.

(84) **S134** can optionally further include reporting and/or recording (e.g., tallying, saving, etc.) one or more sections based on the classification. In some variations, sections having a classification indicating that the user is interacting with a screen of the user device (e.g., reading) carry on to further processes of the method, such as determining a risk score in **S140**. Prior to this, any or all of the classifications can optionally be assessed to predict whether or not the user is actively engaging with the user device during the section (e.g., based on motion data, based on CPU data, etc.). In examples, CPU activity must meet a minimum threshold (e.g., as described above) in order to be marked as active. Additionally or alternatively, the classifications can be otherwise analyzed or not analyzed.

(85) The sections can further optionally be analyzed in relation to a parameter of the vehicle (e.g., speed, as described in **S132**, etc.), such as prior to being reported or recorded. Additionally or alternatively, **S134** can be performed in absence of determining a vehicle parameter and analyzing the sections in light of it.

(86) Additionally or alternatively, the set of tap parameters can be determined in the same way in **S132** and **S134**, and/or be determined in any other suitable way(s).

(87) **S130** is preferably performed multiples times throughout the method **100**, such as throughout the duration of a trip, wherein one of **S132** and **S134** is being implemented at any time point that the user is driving. Additionally or alternatively, **S130** can be performed once, intermittently, in response to a trigger, and/or at any other suitable times.

(88) In one variation, **S130** includes in response to detecting that the device is in a stationary state:

determining a set of tap events occurring in a set of data (e.g., data stream) corresponding to the stationary state, wherein the set of data can include any or all of the information collected in **S110**; determining a set of tap regions based on the set of tap events, wherein determining the set of tap regions includes comparing each of the set of tap events with a maximum temporal spacing threshold (e.g., 10 seconds) and aggregating tap events with a temporal spacing less than the threshold into a tap region; determining a duration of the tap region; eliminating tap regions from further processing which have a duration less than a predetermined threshold (e.g., 3 seconds); optionally determining a processing activity associated with the tap region (e.g., a normalized CPU curve) and moving tap regions to further processing which have a processing parameter above a predetermined threshold (e.g., at least a predetermined number of points above a predetermined threshold value); determining a parameter associated with the vehicle (e.g., speed); and eliminating tap regions from further processing which are associated with a vehicle parameter below a predetermined threshold (e.g., 11 mph). Additionally or alternatively **S132** can include any other suitable processes.

(89) In a specific example of the first variation, determining the set of tap events includes determining a set of rates of gravity deviation and determining a subset (e.g., top 10%) of the rates of gravity deviation having the highest value; aggregating the subset of tap events into tap regions which have a spacing of less than or equal to 10 seconds between adjacent tap events; discarding tap regions from further processing which have a duration of less than 3 seconds; discarding tap regions from further processing which are associated with a CPU activity below a predetermined threshold; and discarding tap regions from further processing which are associated with (e.g., occurring contemporaneously) with a vehicle speed less than or equal to 11 miles per hour.

(90) In a second variation, additional or alternative to the first variation, **S130** includes in response to detecting that the device is in a non-stationary state: splitting data (e.g., data stream), wherein the set of data can include any or all of the information collected in **S11**, corresponding to the non-stationary state into a set of sections based on a set of transitions (e.g., motion transitions, orientation transitions, etc.) occurring in the data; classifying each section (e.g., as a call, as a user interacting with/touching a screen of the user device, as unknown, as shown in FIGS. 4A-4C, etc.) based on orientation information associated with the user device; and optionally reporting and/or otherwise marking classifications associated with the riskiest behavior (e.g., user interacting with/touching a screen). Additionally, **S134** can include discarding sections associated with a vehicle parameter outside of a set of threshold (e.g., below a predetermined speed threshold), and/or otherwise analyzing the data.

(91) 4.4 Method: Determining a Risk Score Based on the Set of Tap Parameters **S140**

(92) The method **100** can optionally include determining a risk score based on the set of tap parameters **S140**, which functions to quantify the riskiness of the user's interactions with his or her user device.

(93) The risk score is preferably determined based on a phone tapping risk score, which is determined based on the percentage of a trip in which the user was tapping on the user device. The risk score can additionally or alternatively be determined based on a phone handling risk score, which is determined based on the percentage of a trip the user was handling the user device. The events corresponding to the user handling the user device preferably includes the duration in time in which the device is mobile. Additionally or alternatively, the handling events can include events wherein the user device is mobile but the user is not interacting with the screen (e.g., when the user is on a call, when the user is not on a call and not interacting with the screen, when the user is holding the user device, etc.), but can additionally or alternatively include any other suitable events.

(94) In variations wherein the risk score takes into account both the phone tapping and phone handling scores, the phone tapping scores and phone handling scores (e.g., all events wherein the user device is mobile) can be directly added together without taking into account overlapping events. This can function to signify that a user device handling event and a phone tapping event

occurring simultaneously is as risky or nearly as risky as them occurring separately. Additionally or alternatively, overlapping events can be taken into account, the risk scores can be weighted prior to combining, a deep learning model and/or one or more algorithms can be used to determine the risk score, and/or the risk score can be otherwise determined.

(95) The risk score can be determined on a per trip basis (e.g., based on vehicle location data, based on vehicle motion data, etc.), aggregated over multiple trips, and/or be otherwise determined and/or associated with at least a portion of a trip and/or multiple trips.

(96) Additionally or alternatively, any or all of the methods, processes, and embodiments described in U.S. application Ser. No. 15/835,284, filed 7 Dec. 2017, which is incorporated herein in its entirety by this reference can be implemented in **S140**.

(97) 4.5 Method: Determining an Output Based on the Risk Score **S150**

(98) The method **100** can optionally include determining an output based on the risk score **S150**, which can function to update a driver assessment, produce an actionable output, and/or perform any other suitable function.

(99) The output can optionally include determining and/or updating a report, such as a report sent to any or all of: an insurance company (e.g., in an event that an accident occurs, to reward a user for safe driving, to determine a user's insurance cost, to determine whether the user could potentially be overpaying with their current policy and/or if the user could potentially pay less with a new insurance policy, to recruit the user to the insurance company, to persuade the user to leave his or her current insurance provider and receive a lower rate at another insurance provider, etc.), a 3.sup.rd party company and/or application, the user (e.g., to facilitate safe driving, to alert the user to potentially risky behavior, to provide the user with actionable feedback indicating which behaviors of theirs specifically contribute the most risk in their driver risk score, etc.), a supervisory agency (e.g., a probation officer), another user (e.g., a family member, a passenger, etc.), and/or any other suitable user. Additionally or alternatively, the output can include any or all of: updating a user profile (e.g., at the primary client application), updating an aggregated risk score (e.g., associated with the user profile), transmitting a notification (e.g., to any of the individuals described above, in response to the risk score exceeding a predetermined threshold, etc.), restricting access to any or all of user device, and/or producing any suitable output.

(100) In some variations, **S250** includes calculating a set of one or more insurance costs for a driver based on a driver risk score, where the insurance costs can be any or all of: provided to the driver (e.g., indicating what he or she would pay at each of a set of one or more insurance providers such that the driver can decide if he or she wants to be insured by a particular insurance provider); provided to the insurance provider (e.g., such that the insurance provider can recruit the driver and/or reach out with a particular insurance offer based on the cost and/or driver risk score, such that the insurance provider can determine a subset of "best" drivers [e.g., those with the lowest driving risk] to recruit and/or provide offers to, such that the insurance provider can price insurance policies accordingly based on a corpus of driver risk scores, etc.); provided to a 3.sup.rd party entity such as a 3.sup.rd party company and/or application (e.g., financial services application, 3.sup.rd party application which matches the driver with a best insurance provider based on its associated insurance cost, etc.); and/or otherwise suitably used.

(101) In a set of specific examples, for instance, a 3.sup.rd party entity (e.g., non-insurance application executing on a user device of a driver) already engaged with the driver can request, prompt, and/or initiate the commencement of a predetermined period of time for which data will be collected in determining a risk score of the driver, where the risk score of the driver during this period can be used to determine whether or not he or she qualifies for a particular insurance offer and/or to determine a particular cost associated with the insurance offer. Alternatively, rather than a period of time, the driving period used in determining the driver risk score can correspond to any or all of: a predetermined number of drivers, a predetermined driving distance, a predetermined amount of driving time, a dynamically determined period, and/or any other time periods.

(102) The method **100** can optionally further include, at any time during the method **100**, classifying (e.g., verifying) a user as the driver, or alternatively a passenger. This is preferably performed based on any or all of the methods, processes, and embodiments described in U.S. application Ser. No. 16/022,120, filed 28 Jun. 2018, which is incorporated herein in its entirety by this reference, but can additionally or alternatively be determined in any other suitable way.

(103) Embodiments of the system and/or method can include every combination and permutation of the various system components and the various method processes, wherein one or more instances of the method and/or processes described herein can be performed asynchronously (e.g., sequentially), concurrently (e.g., in parallel), or in any other suitable order by and/or using one or more instances of the systems, elements, and/or entities described herein.

(104) 4.6 Method: Variations

(105) In a first variation of the method **100** (e.g., as shown in FIG. 5), the method **100** includes any or all of: continuously collecting (e.g., while the user device is in a vehicle, while the user device is in a vehicle as determined by location information, etc.) motion information (e.g., from an accelerometer, from a gyroscope, from an orientation sensor, etc.) associated with the user device, performance information (e.g., processing activity, CPU activity, network transfer information, amount of free memory, etc.) associated with the user device, and optionally handling information (e.g., from a proximity sensor, from an optical sensor, etc.) associated with the user device, which is used to assess a user's interaction with the user device in subsequent processes of the method; collecting location information to determine one or more parameters of the vehicle (e.g., speed, location in space, etc.), which can be used in subsequent processes of the method (e.g., to eliminate data, to appropriately determine a risk score, etc.); classifying, based on any or all of the information collected in **S110** and an algorithm (e.g., of a machine learning model, a phone movement algorithm, etc.), a classification of at least whether the device is in a stationary state or a non-stationary state, which functions to trigger one or more actions in **S130**; in an event that the device is in a stationary state: determining a set of tap events occurring in a set of data (e.g., data stream) corresponding to the stationary state, wherein the set of data can include any or all of the information collected in **S110**; determining a set of tap regions based on the set of tap events, wherein determining the set of tap regions includes comparing each of the set of tap events with a maximum temporal spacing threshold (e.g., 10 seconds) and aggregating tap events with a temporal spacing less than the threshold into a tap region; determining a duration of the tap region; eliminating tap regions from further processing which have a duration less than a predetermined threshold (e.g., 3 seconds); optionally determining a processing activity associated with the tap region (e.g., a normalized CPU curve) and moving tap regions to further processing which have a processing parameter above a predetermined threshold (e.g., at least a predetermined number of points above a predetermined threshold value); determining a parameter associated with the vehicle (e.g., speed); and eliminating tap regions from further processing which are associated with a vehicle parameter below a predetermined threshold (e.g., 11 mph). Additionally or alternatively **S132** can include any other suitable processes; in an event that the device is in a non-stationary state: splitting data (e.g., data stream), wherein the set of data can include any or all of the information collected in **S11**, corresponding to the non-stationary state into a set of sections based on a set of transitions (e.g., motion transitions, orientation transitions, etc.) occurring in the data; classifying each section (e.g., as a call, as a user interacting with/touching a screen of the user device, as unknown, etc.) based on orientation information associated with the user device; and optionally reporting and/or otherwise marking classifications associated with the riskiest behavior (e.g., user interacting with/touching a screen). Additionally, **S134** can include discarding sections associated with a vehicle parameter outside of a set of threshold (e.g., below a predetermined speed threshold), and/or otherwise analyzing the data; determining a phone tapping score based on one or both of **S132** and **S134**; determining a phone handling score based on **S132**; determining a risk score by combining (e.g., directly adding, adding in a weighted fashion, combining according to an

algorithm, etc.) the phone tapping score and the phone handling score; and optionally determining an output (e.g., updating a user profile, updating an insurance model/report, etc.) based on the risk score.

(106) In a second variation of the method **100** (and/or corresponding system), additional or alternative to the first, the method can optionally include any or all of: identifying a user as a driver; determining a current insurance cost (e.g., current price user is paying for automobile insurance) for the driver (e.g., based on driver feedback at an application such as a 3.sup.rd party application, based on information at an application of the user's current insurance provider, etc.); provide a notification to the driver (e.g., in response to determining that the driver is engaging with a 3.sup.rd party application related to car shopping, managing cash flow, financial planning, and/or other activities) which requests their interest in engaging with an insurance provider (e.g., for an insurance package, to switch insurance providers, etc.); initiating the commencement of a data collection period (e.g., predetermined duration of driving time); during the data collection period and/or at any other time(s) (e.g., in absence of a data collection period, prior to a data collection period, etc.), collecting data from a set of sensors onboard a user device; with a set of models and/or algorithms (e.g., machine learning algorithms and/or models, artificial intelligence powered algorithms and/or models, etc.), calculating a set driving behavior parameters (e.g., device usage based on tapping detections, aggressive driving metrics, speed, etc.) based on data collected during the data collection period and/or at any other times; calculating a driver risk score for the driver based on the driving behavior parameters; comparing the driving behavior parameters and/or the driver risk score with a set of thresholds (e.g., requirements) wherein the thresholds can be any or all of: associated with (e.g., specified by) a particular insurance provider, determined independently of a particular insurance provider, etc.; determining a set of outputs (e.g., binary determination of whether or not the driver qualifies for an insurance plan and/or offer, determination of a cost that the insurance provider would charge the user, determination of which of set of insurance providers would offer the driver the lowest cost plan, etc.); triggering an action based on the set of outputs (e.g., transmitting a notification to the driver which optionally includes any or all of the outputs, establishing communication between the driver and an insurance provider, a notification to the user which conveys the decrease in insurance cost which would result in switching insurance providers, etc.); and/or any other processes.

(107) Additionally or alternatively, the method **100** can include any other suitable processes performed in any suitable combination. In some variations, for instance, the method **200** includes updating (e.g., re-training) any or all of the algorithms and/or models utilized in the method based on outputs produced in the method, the inputs used to produce the outputs, feedback received from the user (e.g., acceptance of a new insurance policy, rejection of a new insurance policy, etc.) and/or entity (e.g., 3.sup.rd party application, insurance provider, etc.), and/or any other information.

(108) Additional or alternative embodiments implement the above methods and/or processing modules in non-transitory computer-readable media, storing computer-readable instructions. The instructions can be executed by computer-executable components integrated with the computer-readable medium and/or processing system. The computer-readable medium may include any suitable computer readable media such as RAMs, ROMs, flash memory, EEPROMs, optical devices (CD or DVD), hard drives, floppy drives, non-transitory computer readable media, or any suitable device. The computer-executable component can include a computing system and/or processing system (e.g., including one or more collocated or distributed, remote or local processors) connected to the non-transitory computer-readable medium, such as CPUs, GPUs, TPUS, microprocessors, or ASICs, but the instructions can alternatively or additionally be executed by any suitable dedicated hardware device.

(109) As a person skilled in the art will recognize from the previous detailed description and from

the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

Claims

1. A method for determining and utilizing a driver risk score for a user, the method comprising: detecting that a user has initiated a driving trip in a vehicle; during the driving trip, collecting a set of data at a user device associated with the user, the set of data comprising: acceleration information from a set of accelerometers onboard the user device; orientation information from a gyroscope onboard the user device; and system performance information associated with the user device, wherein the system performance information comprising at least one of: processing activity information, network transfer information, or an amount of free memory; selecting a state of the user device from a set of states based on the set of data, wherein the set of states comprises: a stationary state, wherein in the stationary state, the user device is determined to be stationary relative to the vehicle; a non-stationary state, wherein in the non-stationary state, the user device is determined to be moving relative to the vehicle; in an event that the stationary state is selected, determining a set of tap parameters associated with a touch-sensitive surface of the user device; eliminating a subset of the set of tap parameters from further processing based on a system performance threshold associated with the system performance information; in an event that the non-stationary state is selected, determining a classification of a set of classifications for the user device; calculating the driver risk score based on at least one of: the set of tap parameters or the classification; comparing the driver risk score with a set of thresholds; determining an insurance cost for the user based on the comparison; facilitating provision of a set of outputs determined based on the insurance cost to the user at an application executing on the user device; and suspending usage of the user device during the driving trip based on the comparison.
2. The method of claim 1, wherein each of the subset of tap parameters is associated with a time period in which at least one of a processing activity or a network transfer is below a predetermined threshold.
3. The method of claim 1, further comprising eliminating a subset of the set of tap parameters from further consideration, wherein each of the subset of tap parameters is associated with a time period in which a speed associated with the vehicle is below a predetermined threshold.
4. The method of claim 1, wherein the state is selected based on processing the set of data with a set of machine learning algorithms.
5. The method of claim 4, further comprising updating the set of machine learning algorithms based on the set of data and the set of outputs.
6. The method of claim 5, further comprising processing a second set of data from a second user with the updated set of machine learning algorithms.
7. The method of claim 1, further comprising establishing communication between a set of client applications in response to the comparison, wherein a first client application of the set of client applications is associated with an insurance provider.
8. The method of claim 7, wherein facilitating the provision of the set of outputs is performed in response to establishing communication.
9. The method of claim 1, further comprising determining a set of multiple insurance costs associated with a set of multiple insurance providers, the set of multiple insurance costs comprising the insurance cost, and selecting the insurance cost from the set of multiple insurance costs based on a second comparison between the set of multiple insurance costs.
10. A system for determining and utilizing a driver risk score for a user, the system comprising: a processing subsystem in communication with a first application executing on user device, wherein the processing subsystem: detects that a user has initiated a driving trip in a vehicle; during the driving trip, collects a set of data, the set of data comprising: acceleration information from a set of

accelerometers onboard the user device; orientation information from a gyroscope onboard the user device; and system performance information associated with the user device, comprising at least one of: processing activity information, network transfer information, or an amount of free memory; selects a state of the user device from a set of states based on the set of data, wherein the set of states comprises: a stationary state, wherein in the stationary state, the user device is determined to be stationary relative to the vehicle; a non-stationary state, wherein in the non-stationary state, the user device is determined to be moving relative to the vehicle; in an event that the stationary state is selected, determines a set of tap parameters associated with a touch-sensitive surface of the user device; in an event that the non-stationary state is selected, determines a classification of a set of classifications for the user device; eliminates a subset of the set of tap parameters from further processing based on a system performance threshold associated with the system performance information; calculates the driver risk score based on at least one of the set of tap parameters or the classification; compares the driver risk score with a set of thresholds; determines an insurance parameter for the user based on the comparison facilitates provision of the insurance parameter to the user at the first application; and suspends usage of the user device during the driving trip based on the comparison.

11. The system of claim 10, wherein the processing subsystem is onboard the user device.

12. The system of claim 10, wherein the insurance parameter comprises an insurance cost.

13. The system of claim 12, wherein the processing subsystem further determines a set of multiple insurance costs associated with a set of multiple insurance providers, the set of multiple insurance costs comprising the insurance cost, wherein the processing subsystem selects the insurance cost from the set of multiple insurance costs based on a second comparison between the set of multiple insurance costs.

14. The system of claim 12, wherein the processing subsystem is further in communication with a second application, wherein the insurance cost is received from the second application.

15. The system of claim 14, further comprising establishing communication between the first and second applications in response to comparing the driver risk score with a set of thresholds.

16. The system of claim 15, wherein the set of thresholds is received from the second application.

17. The system of claim 10, wherein the processing subsystem further eliminates a second subset of the set of tap parameters from further consideration, wherein each of the second subset of tap parameters is associated with a time period in which a speed associated with the vehicle is below a threshold.

18. The system of claim 10, wherein the state is selected based on processing the set of data with a set of machine learning algorithms.

19. The method of claim 1, wherein the set of outputs comprises a notification of time spent in a state of distracted driving during the driving trip.
